Projector-Lipschitz Propagation Envelopes

Why Universal Unit Tail Propagation Is False

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Abstract

The exact hybrid budgets of RH-97 show no tail amplification on the archived packet chains: every propagated quotient contribution is no larger than its local loss. We test whether this unit behavior can be promoted to a theorem. It cannot.

We first derive the correct conditional factorization. For rank-r packets U, V and endpoint Gramian G,

$$|\mathcal{E}_G(U) - \mathcal{E}_G(V)| \leq \|G\|_F \, \|UU^* - VV^*\|_F \, .$$

If an adaptive rank-r Ritz packet has capture loss ε relative to the leading rank-r packet of a compressed Hermitian matrix with gap $\gamma = \lambda_r - \lambda_{r+1} > 0$, then

$$\|P_{\mathrm{ad}} - P_{\mathrm{full}}\|_F \leq \sqrt{\frac{2\varepsilon}{\gamma}}.$$

If the future full block is projector-Lipschitz with constant L, these give the conditional projector block envelope

$$|\Delta_{\mathrm{end}}| \leq \|G_N\|_F \, L \sqrt{\frac{2\varepsilon}{\gamma}}.$$

All 38 production omissions generated by the three RH-96 cutoffs satisfy the gap-to-projector, endpoint Lipschitz, and combined conditional bounds. Tail amplification is at most $1 + 4 \times 10^{-16}$ numerically. Projector secant amplification nevertheless reaches 8.85112, showing that endpoint energy is less sensitive than packet geometry.

We then give an explicit 6×6, trace-one, strictly positive two-step projected-cross Ritz counterexample. A width-one first update has validated local loss at most 5.28909×10^{-4} relative to width two. After one common width-two future refresh, the endpoint tail difference is at least 2.35338×10^{-2}, a certified amplification greater than 44.4949. Thus unit propagation is false even for normalized positive Gramians.

The missing route object is not a universal constant one, but a neighborhood-uniform projector-Lipschitz bound for the actual separated reduced maps. No such uniform bound, repeated-block theorem, Hilbert–Polya operator, zero identification, or Riemann Hypothesis result is claimed.

Keywords: spectral projector; Lipschitz propagation; Ritz gap; Grassmann distance; counterexample; validated numerics.

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1 Introduction

RH-97 decomposes the endpoint effect of adaptive weak-mode omissions into exact nonlinear hybrid contributions [4]. On all archived chains, the absolute endpoint contribution of each omission is no larger than its local one-step tail loss, up to roundoff. If this were a structural nonexpansiveness theorem, the replay-free horizon budget would be immediate.

There is reason for caution. A projected-cross Ritz update is not a metric projection in one fixed Hilbert geometry. Its trial space depends on the incoming packet, the Gramian changes with time, and a small local energy difference can correspond to a packet displacement that a later Gramian sees strongly. Ritz monotonicity controls one packet under one Gramian; it does not compare two packet histories.

The paper therefore separates three transformations:

$$\begin{aligned}&\text{local energy loss} \longrightarrow \text{local projector displacement}\\ &\qquad\longrightarrow \text{future projector displacement} \longrightarrow \text{endpoint energy effect.}\end{aligned}$$

The first arrow is controlled by a compressed spectral gap. The last is controlled by the endpoint Gram norm. The middle arrow is the genuinely dynamical object: a projector-Lipschitz constant for the future full block.

This factorization gives a valid conditional theorem. More importantly, a small explicit counterexample proves that the middle factor cannot be silently replaced by one. The production data remain favorable, but their favorable behavior must be proved from the specific map family.

2 Endpoint tail/projector Lipschitz theorem

For an isometric rank-r packet V, let $P_V = VV^*$ and

$$\mathcal{E}_G(V) = \operatorname{tr} G - \operatorname{tr}(GP_V).$$

Theorem 2.1 (Endpoint tail/projector Lipschitz theorem)**.** *For $G = G^*$ and any two rank-r packets U, V,*

$$|\mathcal{E}_G(U) - \mathcal{E}_G(V)| \le \|G\|_F \, \|P_U - P_V\|_F \, . \tag{1}$$

Proof. The trace terms cancel:

$$\mathcal{E}_G(U) - \mathcal{E}_G(V) = \operatorname{tr}\big(G(P_V - P_U)\big).$$

Apply the Frobenius Cauchy–Schwarz inequality. □

The theorem is exact and global. It does not require a spectral gap or a common trial space. Its usefulness depends on controlling the future packet projector distance.

3 Local gap loss-to-projector theorem

Let $H = H^*$ be a compressed Ritz matrix of dimension at least $r + 1$, with eigenvalues

$$\lambda_1 \ge \cdots \ge \lambda_r > \lambda_{r+1} \ge \cdots .$$

Let P_* be its leading rank-r spectral projector and Q any rank-r orthogonal projector in the same compressed space. Define the capture loss

$$\varepsilon = \operatorname{tr}(HP_*) - \operatorname{tr}(HQ) \ge 0.$$

Theorem 3.1 (Local gap loss-to-projector theorem)**.** *With $\gamma = \lambda_r - \lambda_{r+1} > 0$,*

$$\|P_* - Q\|_F \le \sqrt{\frac{2\varepsilon}{\gamma}}. \tag{2}$$

Proof. Work in an eigenbasis of H. If q_i are the diagonal entries of Q in that basis, then $0 \le q_i \le 1$ and $\sum_i q_i = r$. Hence

$$\begin{aligned}\varepsilon &= \sum_{i\le r} \lambda_i(1 - q_i) - \sum_{i>r} \lambda_i q_i\\ &\ge \gamma \sum_{i>r} q_i = \gamma\big(r - \operatorname{tr}(P_*Q)\big).\end{aligned}$$

For rank-r projectors,

$$\|P_* - Q\|_F^2 = 2r - 2\,\mathrm{tr}(P_* Q),$$

which gives (2). This is the elementary sin-theta energy bound behind standard spectral projector perturbation estimates [1, 3]. □

In the adaptive enrichment setting, Q is the adaptive packet embedded in the full-width compressed basis, and P_* is the full-width leading packet. The local quotient loss is exactly ε.

4 Conditional projector block envelope

Let $\mathcal{F}_{j+1:N}$ denote the future composition of full-width packet refresh maps from time $j+1$ to N. Suppose that on a set containing two local outputs,

$$\left\| P_{\mathcal{F}_{j+1:N}(U)} - P_{\mathcal{F}_{j+1:N}(V)} \right\|_F \le L_{j,N} \left\| P_U - P_V \right\|_F . \tag{3}$$

Corollary 4.1 (Conditional projector block envelope)**.** *If the local full-width compressed gap is $\gamma_j > 0$ and the local adaptive capture loss is ε_j, then*

$$|\Delta_{j,N}| \le \|G_N\|_F \, L_{j,N} \sqrt{\frac{2\varepsilon_j}{\gamma_j}}. \tag{4}$$

Proof. Apply Theorem 3.1 at time j, then (3), and finally Theorem 2.1. □

The bound identifies the missing constant precisely. A pairwise secant value

$$L_{j,N}^{\mathrm{sec}} = \frac{\left\|P_{\mathcal{F}(U)} - P_{\mathcal{F}(V)}\right\|_F}{\|P_U - P_V\|_F}$$

certifies one concrete pair a posteriori. A replay-free theorem needs a uniform $L_{j,N}$ on a neighborhood containing all admissible quotient outputs.

5 Universal unit propagation is false

We now reject the tempting assertion

$$|\Delta_{j,N}| \le \varepsilon_j$$

for arbitrary normalized positive Gram sequences.

Proposition 5.1 (Strictly positive two-step counterexample)**.** *There exist 6×6 real symmetric positive definite matrices G_1, G_2 with $\mathrm{tr}\, G_1 = \mathrm{tr}\, G_2 = 1$, and a rank-two initial packet V, such that:*

1. *at time one, width one has positive tail loss ε relative to width two;*
2. *applying the same width-two refresh under G_2 to both outputs gives endpoint tail difference Δ satisfying*

$$\frac{|\Delta|}{\varepsilon} > 44.4949.$$

Proof. The matrices and packet are archived as exact binary inputs in the audit record. Their smallest binary64 eigenvalue lower diagnostics are 5.55×10^{-3} and 1.44×10^{-7}, respectively, and both traces are exactly one in binary64.

At 384-bit Arb precision, the local loss has outward upper bound

$$\varepsilon \leq 5.2890877381 \times 10^{-4},$$

while the absolute endpoint effect has outward lower bound

$$|\Delta| \geq 2.3533757224 \times 10^{-2}.$$

Division of the outward bounds gives $|\Delta|/\varepsilon > 44.49492689$. The endpoint tail/projector Lipschitz bound remains green, so the amplification is genuine and not a failed numerical inequality. □

Remark 5.2. The counterexample is not a claim about the production Gaussian family. It shows that positivity, trace normalization, and projected-cross Ritz structure alone do not imply unit propagation.

6 Production audit

We replay all omissions generated by the thresholds $10^{-8}, 10^{-6}, 10^{-4}$ on the ten RH-96 channels, giving 38 local/future pairs. For each pair we record:

1. the local adaptive/full capture loss and full compressed gap;
2. the actual local projector distance and the bound (2);
3. the future full-block endpoint projector distance and secant multiplier;
4. the endpoint tail effect and bound (1);
5. the combined conditional bound (4), using the pairwise secant.

Quadratic tail forms are evaluated from exact binary packets in Arb at 384 bits [2].

All 38 local gap-distance bounds close. All 38 endpoint tail/projector bounds close. All 38 conditional secant envelopes close. The largest gap-distance slack factor is 19.29.

Tail amplification is numerically at most

$$1.0000000000000004,$$

so all 38 pairs are unit-propagating within a 10^{-12} roundoff allowance. The strict outward comparison is inconclusive at fourteen near-equality cases, mostly omissions at the endpoint itself.

Packet geometry behaves differently. The largest projector secant multiplier is

$$8.85111975$$

for the primary threshold, and the 10^{-6} family reaches 7.75018. Future refresh can amplify Grassmann displacement even while the endpoint tail functional suppresses its energetic effect.

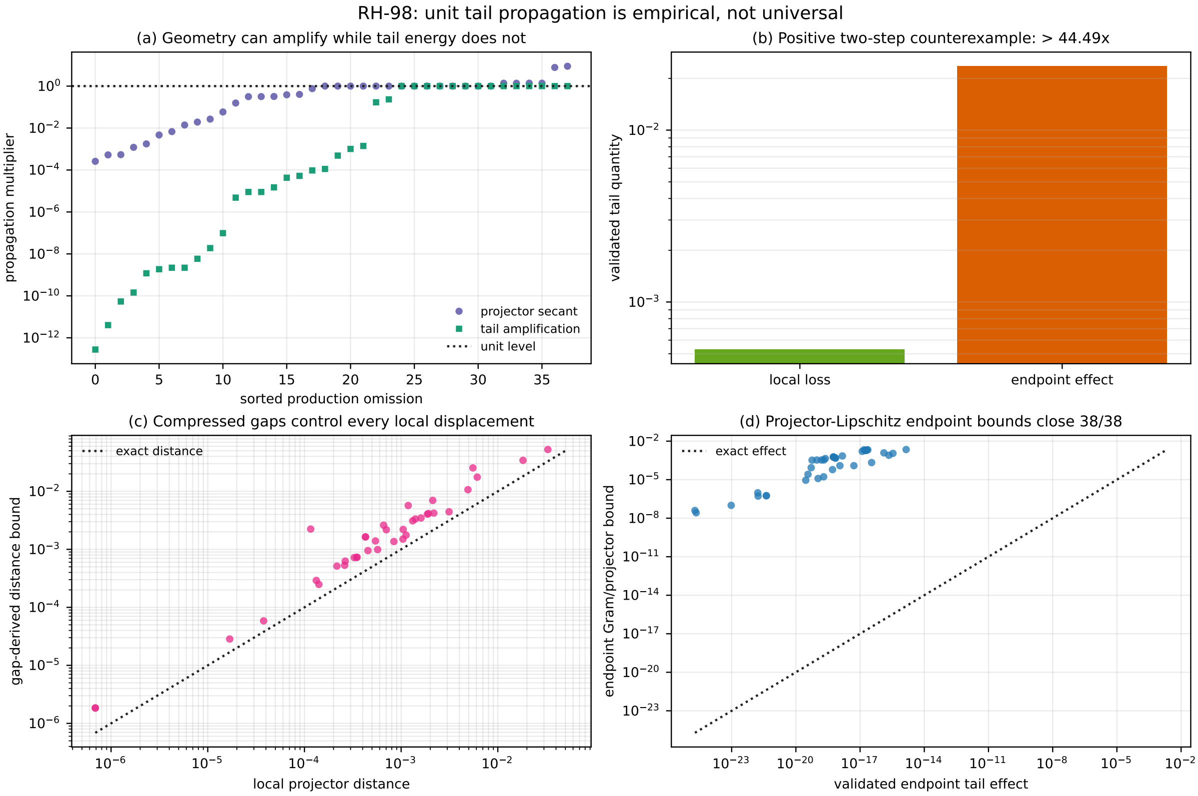

Figure 1: Production pairs satisfy the conditional projector envelope, but projector secants can exceed eight. The explicit positive counterexample rejects universal unit tail propagation by a factor greater than forty-four.

7 Next gate: a differential projector tube

The correct RH-99 target is a neighborhood-uniform version of (3). For a separated spectral projector, its Fréchet derivative is represented by a reduced resolvent or Sylvester equation. One should differentiate the complete projected-cross Ritz map, including:

- the incoming packet projector;
- the projected cross operator and selected singular subspace;
- the enriched compressed matrix;
- the leading compressed Ritz projector.

If each selected singular and Ritz cluster remains separated on a verified projector ball, derivative norms can be integrated to obtain a genuine local Lipschitz constant. Branch changes or gap collapse would instead provide a precise negative result.

8 Claim boundary

The endpoint tail/projector Lipschitz theorem, local gap loss-to-projector theorem, conditional projector block envelope, and positive-definite counterexample are exact finite-dimensional results. The production secants are pairwise a posteriori diagnostics. We do not prove a neighborhood-uniform refresh Lipschitz constant, a replay-free uniform block envelope, repeated-block contraction, Stage-A closure, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

9 Conclusion

Unit propagation is false. The archived family happens to show unit tail behavior, but a normalized positive two-step Ritz system can amplify local loss by more than forty-four. The valid route is projector-metric: compressed gaps convert loss to displacement, future map Lipschitz constants propagate displacement, and the endpoint Gram converts it back to energy.

All production pairs satisfy this factorized envelope. The next paper must turn their secant constants into verified differential tubes.

References

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